

COST SUMMARY · MONTHLY RUNNING INFRASTRUCTURE

Infrastructure Cost Breakdown

Monthly cloud + third-party service costs for running CEFANET CDF-MS at three scale points. All figures in USD per month with Zambian Kwacha equivalents at K25 / USD. Costs exclude one-time setup, labour, training and field hardware.

MVP SCALE

\$345 / mo

5 constituencies · K8,625

PILOT SCALE

\$890 / mo

10 constituencies · K22,250

NATIONAL SCALE

\$3,100-\$6,200 / mo

226 constituencies · K78k-K155k

LINE ITEMS

Service-by-service monthly cost

SERVICE	MVP \$	PILOT \$	NATIONAL \$	TIER
AWS ECS Fargate (API + workers)	80	150	600 - 900	Compute
AWS RDS PostgreSQL Multi-AZ	90	140	500 - 700	Data
AWS ElastiCache Redis	30	50	180 - 280	Data
AWS S3 + CloudFront (photos, reports)	20	60	360 - 720	Storage
AWS Cognito (identity, MFA)	0	0	80 - 200	Auth
AWS Secrets Manager + KMS	5	10	30	Security
AWS Route 53 + ACM + WAF	25	30	90	Edge
AWS SES (transactional email)	5	10	60 - 150	Comms
AWS CloudWatch + X-Ray	20	40	170 - 290	Observability
Sentry (error tracking)	26	26	100 - 200	Observability
KoboToolbox hosted (form server)	0	200	200 - 700	Field data
Africa's Talking SMS (variable usage)	10	80	600 - 1,700	Field data
Africa's Talking USSD code rental	0	50	50 - 100	Field data
GitHub Team plan	4	4	4	Dev tools
Domain + uptime monitoring	30	40	90	Edge
TOTAL MONTHLY	~345	~890	3,100 - 6,200	

ANNUALISED

Year-1 running cost projections

PHASE	USD / YEAR	K / YEAR	NOTES	TIER
MVP (Phase 1 deployment)	\$4,140	K103,500	Months 1-6, 5 constituencies	Year-1
Pilot (Phase 2)	\$10,680	K267,000	Months 7-12, 10 constituencies, full features	Year-1
Hardening (Phase 3)	\$15,000	K375,000	Months 13-18, performance work, mobile app live	Year-1
National (Phase 4 + Year-2)	\$37,200 - \$74,400	K930k-K1.86M	Months 19-24, 226 constituencies live	Year-1+

COST LEVERS

How to reduce monthly spend at scale

- > 1-year AWS Reserved Instances on Fargate + RDS reduce compute and database cost by 30-40 percent.
- > S3 Intelligent-Tiering + lifecycle policy moving photos older than 24 months to Glacier reduces storage cost by ~50 percent at national scale.
- > Africa's Talking volume discount activates above 500,000 SMS per year (typical national volume).
- > Self-hosting Kobo on the same AWS account saves ~\$200/mo at the cost of ~0.5 DevOps days/month.



-> Cloudflare in front of CloudFront caches the public dashboard; reduces CloudFront egress costs by 60-70 percent.

ASSUMPTIONS & EXCLUSIONS

What this number does and does not cover

INCLUDED: AWS af-south-1 (Cape Town) compute, database, storage, edge and observability. Identity (Cognito), secrets, certificates, email and SMS gateways. Error tracking, uptime monitoring, version control and CI/CD platforms. KoboToolbox and Africa's Talking subscriptions. Domain registration and renewals.

EXCLUDED FROM THIS SHEET: developer and field-officer salaries; one-time costs (DPIA, penetration testing, branding, legal review); training rounds and per-diems; field hardware (tablets, power banks); communications campaigns; M&E evaluation. See DEVELOPMENT_BUDGET.md for the full breakdown.

FX assumption: K25 per USD. Costs adjust automatically with the prevailing rate.

Prepared for the CEFANET-MLGRD MoU process. Figures revised as live vendor quotes are received. This document is internal and not for public circulation.